Azure IoT Hub Technical Details

Agenda

1. Maximum Messages Per Second
2. Device Update for IoT & Defender for IoT
3. Custom CA Certificates
4. Device Connection Status Monitoring
5. Metrics and API Access
6. Tier Selection & Limitations
7. Pricing Calculation

1. Maximum Messages Per Second

Throttling Limits by Tier

Throttling Limit	Free, B1, and S1	B2 and S2	B3 and S3
Device-to-cloud sends	Higher of 100/sec or 12/sec/unit	120/sec/unit	6,000/sec/unit
Cloud-to-device sends	1.67/sec/unit (100/min/unit)	1.67/sec/unit (100/min/unit)	83.33/sec/unit (5,000/min/unit)

> **Key Points:**
>
> - Limits apply per hub (e.g., S3 tier with 3 units = 18,000 msgs/sec)
> - Azure IoT Hub accepts requests above throttling limit temporarily
> - Continued violations result in queue filling up
> - Requests eventually rejected with `429 ThrottlingException`
> - Plan capacity based on peak message rates, not just averages

2. Device Update for IoT & Defender for IoT

Device Update for IoT

Key Features & Use Cases:

- **Over-the-air (OTA) updates** for IoT devices
- **Security Updates:** Quickly distribute security patches
- **Feature Updates:** Add new functionality to devices
- **Firmware Updates:** Update device firmware
- **Bulk Updates:** Update multiple devices simultaneously
- **End-to-end platform** for publishing, distributing, and managing updates
- Supports everything from tiny sensors to gateways

Defender for IoT

Key Features & Use Cases:

- **Real-time Threat Detection:** Identify threats as they occur
- **Security Alerts:** Prevent potential security breaches
- **Device Monitoring:** Track security status of IoT devices
- **Vulnerability Management:** Identify and address vulnerabilities
- **Comprehensive protection** for Azure-based IoT deployments
- Cloud-based security service with minimal device impact

3. Custom CA Certificates

Automatic Connection Process

1. **Upload CA Certificate:** Add root or subordinate CA to IoT Hub
2. **Proof of Possession:** Verify ownership with verification certificate
3. **Create Device Certificates:** Generate X.509 certs using your CA
4. **Device Registration:** Register devices with IoT Hub
5. **Automatic Connection:** Devices connect using X.509 certificates

Automation Options

> Create automated processes for certificate management:
>
> - Use **Azure Functions** or **Logic Apps** to:
>   - Automatically register new device certificates
>   - Associate certificates with relevant policies
>   - Manage certificate lifecycle
>   - Handle certificate rotation
>   - Implement certificate revocation

4. Device Connection Status Monitoring

Available Methods

Method 1: Device Twin connectionState

- Each device identity has **connectionState** property
- Values: **connected** or **disconnected**
- Only updated for MQTT or AMQP protocols
- Updates may be delayed up to 5 minutes
- Based on protocol-level pings

Method 2: Event Grid (Recommended)

- Publishes **deviceConnected** and **deviceDisconnected** events
- Low-cost and easy to implement
- Checks device status every 60 seconds
- Only publishes most recent state on change
- Reports may be delayed up to 1 minute

Method 3: Custom Heartbeat Pattern

- Devices send messages at specific intervals
- Empty messages can serve as heartbeats
- Service tracks last heartbeat received
- Missing heartbeats indicate potential device issues
- Fully customizable timing and response logic

5. Metrics and API Access

Available Metrics

- **Connected Devices:** Number of devices connected
- **Message Metrics:** Count, size, latency of messages
- **Throttling Metrics:** Number of throttling errors
- **Device Operations:** Registry operations, twin updates, direct methods
- **Routing Metrics:** Message routing success/failure rates
- **Authentication:** Success/failure of authentication attempts

Accessing Metrics

Method 1: Azure Portal

- View metrics from IoT Hub's Metrics section
- Built-in dashboard included with Azure subscription
- No additional fee required
- Create custom charts and pin to dashboards
- Set up alerts based on metric thresholds

Method 2: Azure Monitor API

- Programmatically access IoT Hub metrics
- Query, analyze, and visualize metric data
- Free up to certain usage amount
- Additional fees for high-volume queries or long-term storage

- Data stored for 93 days by default

6. Accessing Metrics via Monitor API

Authentication & Endpoint

```
https://management.azure.com/subscriptions/{subscription-id}/resourceGroups/{re
```

C# Sample Code

```
// Authenticate
var credentials = await ApplicationTokenProvider.LoginSilentAsync(
    tenantId, clientId, clientSecret);

// Create Monitor client
var monitorClient = new MonitorClient(credentials);
monitorClient.SubscriptionId = "your-subscription-id";

// Query metrics
var metrics = await monitorClient.Metrics.ListAsync(
    resourceUri,
    timespan: "PT1H",
    interval: "PT5M",
    metricnames: "d2c.telemetry.ingress.success,connectedDeviceCount",
    aggregation: "Total,Average,Count");
```

Python Sample Code

```
# Authenticate
credential = ClientSecretCredential(
    tenant_id="your-tenant-id",
    client_id="your-client-id",
    client_secret="your-client-secret"
)

# Create metrics client
metrics_client = MetricsClient(credential)
```

```
# Query metrics
response = metrics_client.query_resource(
    resource_uri,
    ["d2c.telemetry.ingress.success", "connectedDeviceCount"],
    timespan={"duration": "PT1H"},
    granularity="PT5M",
    aggregation=["Total", "Average"]
)
```

7. Tier Selection for Your Scenario

Message Volume Calculation

- Year 1: 50,000 devices × 200 messages/day = 10,000,000 messages/day
- Year 2: 100,000 devices × 200 messages/day = 20,000,000 messages/day

Tier Analysis

S1 Tier Analysis

- 400,000 messages/day/unit
- Year 1: 25 units needed
- Year 2: 50 units needed
- Throttling: 300 msgs/sec (Year 1), 600 msgs/sec (Year 2)
- Easily supports 100,000 devices

S2 Tier Analysis

- 6,000,000 messages/day/unit
- Year 1: 2 units needed
- Year 2: 4 units needed
- Throttling: 240 msgs/sec (Year 1), 480 msgs/sec (Year 2)
- Easily supports 100,000 devices

S3 Tier Analysis

- 300,000,000 messages/day/unit
- Year 1 & 2: 1 unit sufficient
- Throttling: 6,000 msgs/sec
- Easily supports 100,000 devices
- Highest performance but most expensive

Potential Limitations

Message Burst Scenarios

- If 10% of devices (5,000 in Year 1) send messages simultaneously
- 5,000 messages at once exceeds S1 and S2 throttling limits
- S3 tier would handle this scenario

Message Size & Operations

- Calculations assume standard message sizes (4KB)
- Additional operations (twins, methods) require more capacity

8. Pricing Calculation

S1 Tier

- Year 1 (25 units): $625/month = $7,500/year
- Year 2 (50 units): $1,250/month = $15,000/year
- Total 2-year cost: $22,500

S2 Tier

- Year 1 (2 units): $500/month = $6,000/year
- Year 2 (4 units): $1,000/month = $12,000/year
- Total 2-year cost: $18,000

S3 Tier

- Year 1 & 2 (1 unit): $2,500/month = $30,000/year
- Total 2-year cost: $60,000

Recommendation

S2 Tier is Most Cost-Effective

- **Sufficient capacity for your message volume**
- **Good balance between cost and performance**
- **Room for growth within each unit**
- **Total 2-year cost: $18,000**
- **Significantly more cost-effective than S1 ($22,500) and S3 ($60,000)**